Home Assignment – Week 1 Day2

1. Try prompting using TL;DR and ELI-15 and observe the difference in output.

Explain what is artificial intelligence

At its core, **Artificial Intelligence (AI)** is a field of computer science dedicated to creating systems capable of performing tasks that typically require human intelligence. These tasks include things like recognizing speech, making decisions, translating languages, and identifying patterns.

While traditional software follows a strict set of "if-then" rules written by programmers, AI uses algorithms to learn from data, allowing it to improve its performance over time without being explicitly programmed for every specific scenario.

Core Components of AI

To understand how AI works, it helps to look at its primary building blocks:

- **Machine Learning (ML):** This is the engine behind most modern AI. It involves feeding a system vast amounts of data so it can identify patterns and make predictions.
- **Neural Networks:** Inspired by the human brain, these are layers of interconnected "nodes" that process information. They are the foundation of **Deep Learning**, which allows AI to handle complex data like photos and videos.
- **Natural Language Processing (NLP):** This branch allows machines to understand, interpret, and generate human language (the technology behind chatbots and voice assistants).
- **Computer Vision:** This enables AI to "see" and interpret visual information from the world,

The Two Main Types of AI

AI is generally categorized into two levels based on its capabilities:

1. **Narrow AI (Weak AI):** This is the AI we use today. It is designed to perform a specific task, such as recommending a song, filtering spam emails, or playing chess. It is "intelligent" only within its predefined lane.
2. **General AI (Strong AI):** This is a theoretical form of AI that would possess the ability to understand, learn, and apply intelligence across any intellectual task, much like a human. We have not yet reached this level of technology.

How AI Learns

Most AI systems learn through one of three primary methods:

How AI Learns

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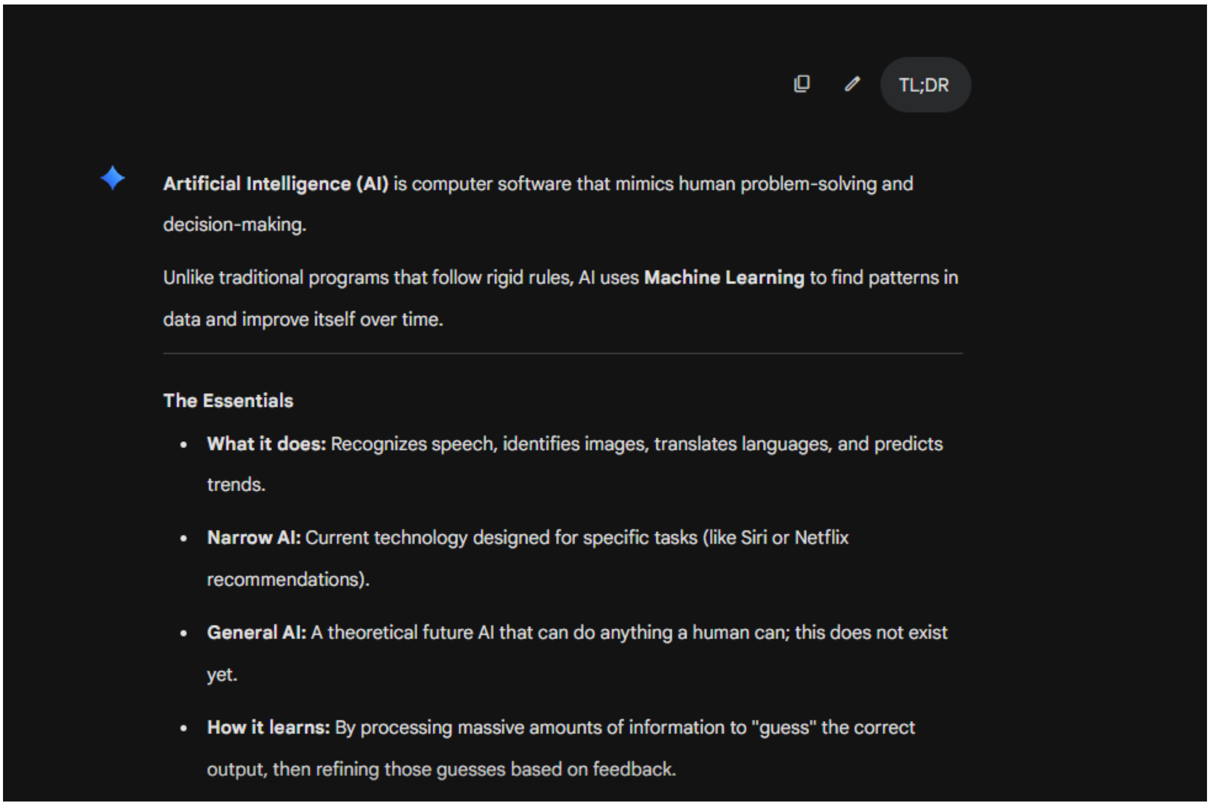
Method	How it Works	Example
Supervised Learning	The AI is trained on labeled data (e.g., "This is a photo of a cat").	Email spam filters
Unsupervised Learning	The AI looks for hidden patterns in unlabeled data.	Customer market segmentation
Reinforcement Learning	The AI learns through trial and error, receiving "rewards" for correct actions.	AI learning to play video games

AI in Daily Life

AI isn't just a futuristic concept; it's already integrated into many everyday tools:

- **Predictive Text:** Guessing the next word you want to type.
- **Navigation:** Apps like Google Maps calculating the fastest route based on real-time traffic.
- **Streaming Services:** Algorithms suggesting movies based on your viewing history.
- **Facial Recognition:** Unlocking your smartphone using your biometrics.

After TL DR


TL;DR

Artificial Intelligence (AI) is computer software that mimics human problem-solving and decision-making.

Unlike traditional programs that follow rigid rules, AI uses **Machine Learning** to find patterns in data and improve itself over time.

The Essentials

- **What it does:** Recognizes speech, identifies images, translates languages, and predicts trends.
- **Narrow AI:** Current technology designed for specific tasks (like Siri or Netflix recommendations).
- **General AI:** A theoretical future AI that can do anything a human can; this does not exist yet.
- **How it learns:** By processing massive amounts of information to "guess" the correct output, then refining those guesses based on feedback.

ELI- 5

ELI-5

Think of **AI** like a robot brain that can learn things just like you do.

Usually, computers are like a recipe book: they only do exactly what the instructions say. But AI is more like a **student**.

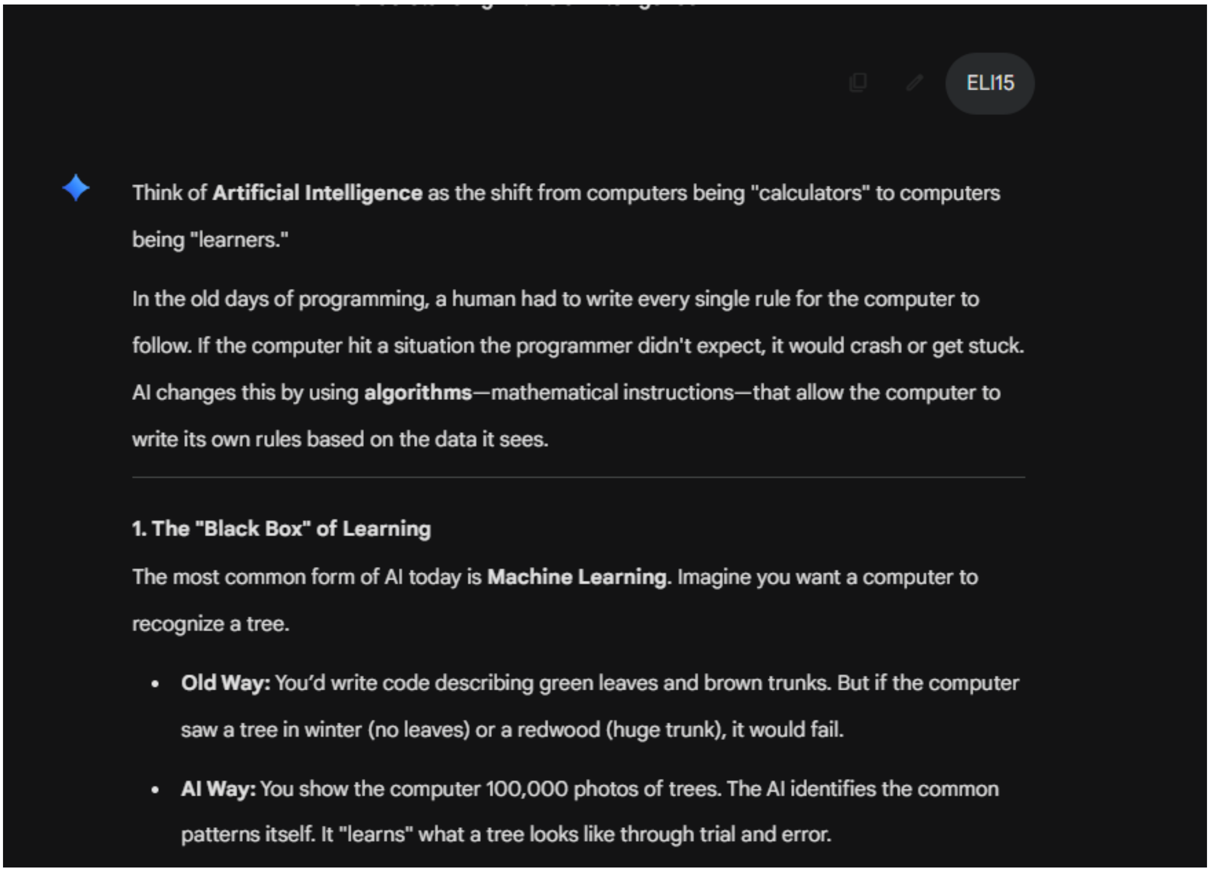
How it Works

- **The Big Library:** Imagine showing a student thousands of pictures of cats. Eventually, that student can look at a brand-new picture and say, "That's a cat!" even if they've never seen that specific one before. That is how AI "learns."
- **Practice makes perfect:** At first, the AI might mistake a dog for a cat. But when we tell it, "No, that's a dog," it remembers that and gets smarter for next time.

AI in Your World

- **Video Games:** It's the "bad guys" who know how to hide or chase you.
- **Talking Speakers:** It's how Alexa or Siri understands what you're saying.

ELI-15



ELI15

Think of **Artificial Intelligence** as the shift from computers being "calculators" to computers being "learners."

In the old days of programming, a human had to write every single rule for the computer to follow. If the computer hit a situation the programmer didn't expect, it would crash or get stuck. AI changes this by using **algorithms**—mathematical instructions—that allow the computer to write its own rules based on the data it sees.

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1. The "Black Box" of Learning

The most common form of AI today is **Machine Learning**. Imagine you want a computer to recognize a tree.

- **Old Way:** You'd write code describing green leaves and brown trunks. But if the computer saw a tree in winter (no leaves) or a redwood (huge trunk), it would fail.
- **AI Way:** You show the computer 100,000 photos of trees. The AI identifies the common patterns itself. It "learns" what a tree looks like through trial and error.

2. Use structured reasoning / step-by-step prompting to generate:

a) Automation Framework Architecture

b) Test Strategy Document

Automation Framework Architecture for salesforce account page

Context

Act as an Automation Architect to design the framework for Salesforce Account creation page with TestNG framework

Instructions
[MANDATORY] Ask me 5 multiple choice questions one by one for the Account standard fields
[MANDATORY] Once after answering the first question ,proceed with net
[MANDATORY] Generate Framework Structure FirstAfter completing all teh questions
[STRICT] Proceed Step by Step for framework implementation
[STRICT] Up on my approval proceed with the next step

Show less

Step 1 — Framework Structure for Salesforce Account Creation Automation (TestNG)

Proposed Hybrid Automation Framework Structure:

```
SalesforceAccountAutomation/
|
├── src/test/java
|   ├── base/
|   |   ├── BaseTest.java
|   |   ├── DriverFactory.java
|   |   └── TestListener.java
|   |
|   ├── pages/
|   |   ├── LoginPage.java
|   |   ├── HomePage.java
|   |   └── AccountPage.java
|   |
```

Ask anything

```
|   |   ├── ConfigReader.java
|   |   ├── ExcelUtil.java
|   |   ├── WaitUtil.java
|   |   ├── ScreenshotUtil.java
|   |   └── RetryAnalyzer.java
|   |
|   └── constants/
|       └── FrameworkConstants.java
|
├── src/test/resources
|   ├── config/
|   |   └── config.properties
|   |
|   ├── testdata/
|   |   └── AccountData.xlsx
|   |
|   └── locators/
|       └── accountpage.properties
|
├── reports/
|
├── screenshots/
|
├── test-output/
|
├── pom.xml
|
└── testng.xml
```

Framework Design Pattern

- Selenium WebDriver
- TestNG
- Page Object Model (POM)
- Data Driven Framework
- Extent Reports
- Maven Build Management
- Retry Mechanism
- Screenshot Capture on Failure

Question 1 of 5 — Account Standard Field

Which Account standard field should be validated as **Mandatory** during Account creation?

Options

A. Account Name
B. Fax
C. Website
D. Description

A. Account Name

Recorded.

- Mandatory Standard Field: **Account Name**

Question 2 of 5 — Account Type Field

Which validation approach should be automated for the **Type** dropdown field?

Options

A. Verify dropdown is displayed only
B. Verify all dropdown values
C. Verify default selected value
D. Verify dropdown allows custom text input

Question 3 of 5 — Phone Field

Which validation should be automated for the **Phone** field?

Options

A. Accept only alphabets
B. Accept special characters only
C. Accept valid numeric phone format
D. Field should be read-only

B. Accept special characters only

Recorded.

- Phone Field Validation: **Accept special characters only**

Question 4 of 5 — Website Field

Which validation should be automated for the **Website** field?

Options

A. Verify URL format validation
B. Verify numeric values only
C. Verify field is hidden
D. Verify field accepts emojis

Question 5 of 5 — Account Ownership Field

Which validation should be automated for the **Account Owner** field?

Options

A. Verify auto-populated logged-in user

B. Verify field accepts multiple owners

C. Verify owner field is disabled permanently

D. Verify owner field accepts numeric values only

B. Verify field accepts multiple owners

Recorded.

- Account Owner Validation: **Verify field accepts multiple owners**

Step 2 — Finalized Automation Scope

Standard Fields Covered

1. Account Name → Mandatory validation
2. Type → Dropdown visibility validation
3. Phone → Special character validation
4. Website → URL format validation

Step 3 — Recommended Tech Stack

Layer	Technology
Language	Java
UI Automation	Selenium WebDriver
Framework	TestNG
Build Tool	Maven
Reporting	Extent Reports
Logging	Log4j
Design Pattern	Page Object Model
Data Handling	Apache POI
CI/CD	Jenkins
Version Control	Git

Step 4 — High Level Framework Flow

```
TestNG XML
    ↓
BaseTest Initialization
    ↓
DriverFactory
    ↓
LoginPage
    ↓
HomePage
    ↓
```

Ask anything

Step 5 — Page Object Classes Planned

Class Name	Responsibility
BaseTest	Browser setup and teardown
DriverFactory	WebDriver initialization
LoginPage	Salesforce login
HomePage	Salesforce navigation
AccountPage	Account creation actions
ConfigReader	Read configuration
ExcelUtil	Test data handling
ScreenshotUtil	Capture screenshots
TestListener	Reporting and logging

Step 6 — Maven Dependencies Planned

XML

```xml
<dependencies>

    <dependency>
        <groupId>org.seleniumhq.selenium</groupId>
        <artifactId>selenium-java</artifactId>
        <version>4.21.0</version>
```

Ask anything

```xml
    <dependency>
        <groupId>org.apache.poi</groupId>
        <artifactId>poi-ooxml</artifactId>
        <version>5.2.5</version>
    </dependency>

</dependencies>
```

Step 7 — Initial Test Scenarios Planned

TC ID	Scenario
ACC_001	Verify Account Name mandatory validation
ACC_002	Verify Type dropdown visibility
ACC_003	Verify Phone field accepts special characters
ACC_004	Verify Website URL validation
ACC_005	Verify Account Owner multiple owner validation

Please approve to proceed with:

Step 8 — Base Framework Implementation (pom.xml + BaseTest + DriverFactory)

Sources

Step 8 — Base Framework Implementation (pom.xml + BaseTest + DriverFactory)

Sources

Proceed

Step 8 — Base Framework Implementation

8.1 pom.xml

XML

```xml
<project xmlns="http://maven.apache.org/POM/4.0.0"
         xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
         xsi:schemaLocation="http://maven.apache.org/POM/4.0.0
         http://maven.apache.org/xsd/maven-4.0.0.xsd">

    <modelVersion>4.0.0</modelVersion>

    <groupId>com.salesforce.automation</groupId>
    <artifactId>SalesforceAccountAutomation</artifactId>
    <version>1.0</version>

    <properties>
        <maven.compiler.source>17</maven.compiler.source>
        <maven.compiler.target>17</maven.compiler.target>
    </properties>

    <dependencies>

        <!-- Selenium -->
        <dependency>
            <groupId>org.seleniumhq.selenium</groupId>
            <artifactId>selenium-java</artifactId>
            <version>4.21.0</version>
```

Ask anything

8.2 DriverFactory.java

Location:

```
src/test/java/base/DriverFactory.java
```

Java

```java
package base;

import io.github.bonigarcia.wdm.WebDriverManager;
import org.openqa.selenium.WebDriver;
import org.openqa.selenium.chrome.ChromeDriver;

import java.time.Duration;

public class DriverFactory {

    private static ThreadLocal<WebDriver> driver = new ThreadLocal<>();

    public static void initDriver() {

        WebDriverManager.chromedriver().setup();

        driver.set(new ChromeDriver());

        getDriver().manage().window().maximize();

        getDriver().manage().timeouts()
                .implicitlyWait(Duration.ofSeconds(10));
    }

    public static WebDriver getDriver() {
        return driver.get();
    }
```

Ask anything

Please approve to proceed with:

Step 10 — Page Object Model Implementation (LoginPage + HomePage + AccountPage)

Approved

Step 10 — Page Object Model (POM) Implementation

10.1 LoginPage.java

Location:

```
src/test/java/pages/LoginPage.java
```

Java

Ask anything

Please approve to proceed with:

Step 13 — Advanced Enterprise Framework Enhancements

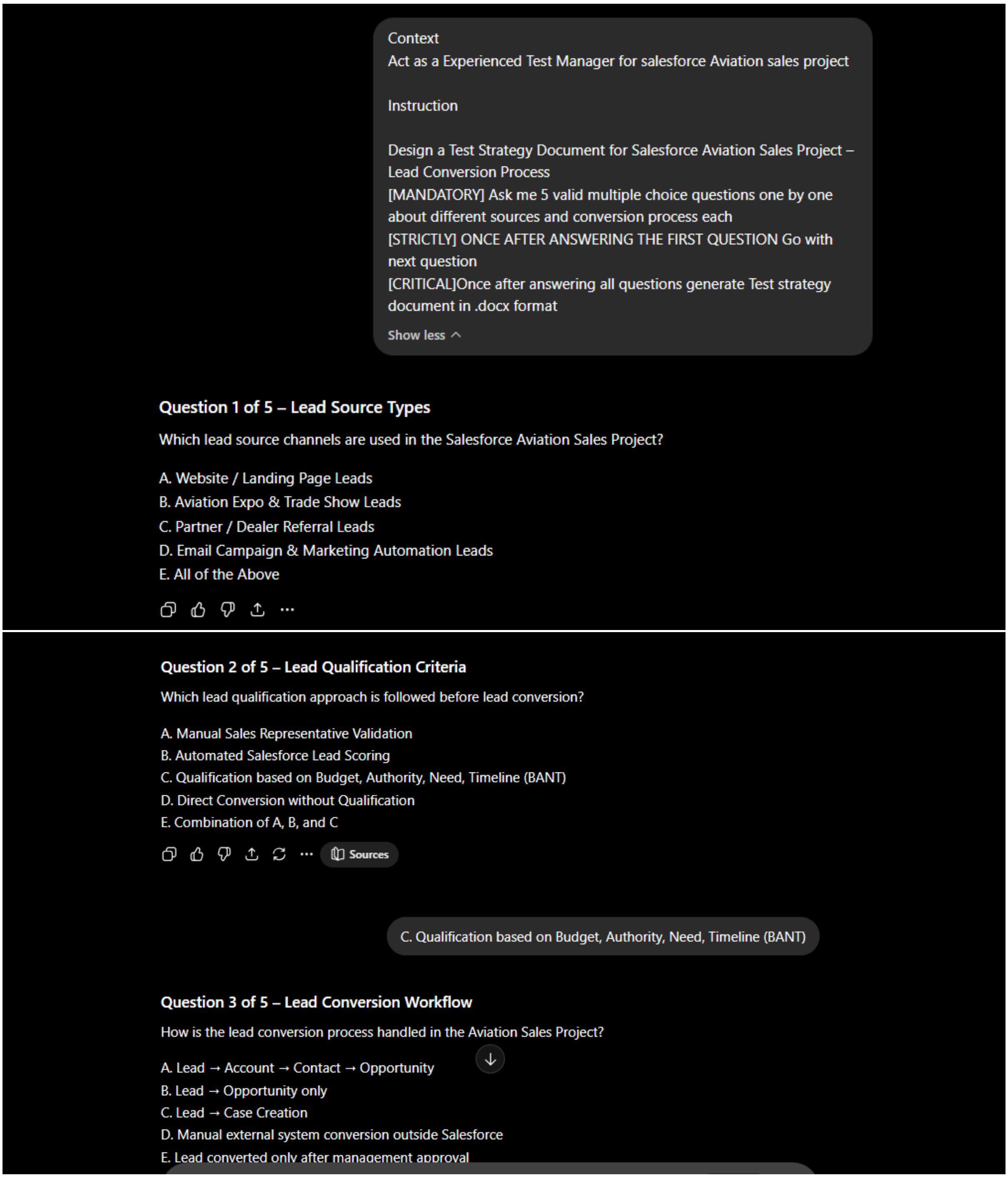
n

Understood.
Current framework implementation is completed up to:

- Base Framework Setup
- POM Design
- Utility Layer
- Reporting Integration
- Data-Driven Testing
- Retry Mechanism
- Screenshot Capture
- End-to-End Account Creation Flow

Framework is now ready for execution and further customization based on project needs.

b) Test Strategy Document for salesforce Aviation project



Context
Act as a Experienced Test Manager for salesforce Aviation sales project

Instruction

Design a Test Strategy Document for Salesforce Aviation Sales Project – Lead Conversion Process
[MANDATORY] Ask me 5 valid multiple choice questions one by one about different sources and conversion process each
[STRICTLY] ONCE AFTER ANSWERING THE FIRST QUESTION Go with next question
[CRITICAL]Once after answering all questions generate Test strategy document in .docx format

Show less

Question 1 of 5 – Lead Source Types

Which lead source channels are used in the Salesforce Aviation Sales Project?

A. Website / Landing Page Leads
B. Aviation Expo & Trade Show Leads
C. Partner / Dealer Referral Leads
D. Email Campaign & Marketing Automation Leads
E. All of the Above

Question 2 of 5 – Lead Qualification Criteria

Which lead qualification approach is followed before lead conversion?

A. Manual Sales Representative Validation
B. Automated Salesforce Lead Scoring
C. Qualification based on Budget, Authority, Need, Timeline (BANT)
D. Direct Conversion without Qualification
E. Combination of A, B, and C

Sources

C. Qualification based on Budget, Authority, Need, Timeline (BANT)

Question 3 of 5 – Lead Conversion Workflow

How is the lead conversion process handled in the Aviation Sales Project?

A. Lead → Account → Contact → Opportunity
B. Lead → Opportunity only
C. Lead → Case Creation
D. Manual external system conversion outside Salesforce
E. Lead converted only after management approval

A. Lead → Account → Contact → Opportunity

Question 4 of 5 – Duplicate Lead Handling

How should duplicate leads be managed during lead conversion?

A. Allow duplicate lead conversion without validation
B. Salesforce Duplicate Rules and Matching Rules enabled
C. Manual duplicate checking by sales team
D. Duplicate leads automatically merged
E. Duplicate validation integrated with external MDM system

Sources

C. Manual duplicate checking by sales team

Question 5 of 5 – Post Conversion Validation

Which validations are mandatory after successful lead conversion?

A. Account, Contact, and Opportunity creation validation
B. Ownership and role assignment validation
C. Email notification and workflow trigger validation
D. Audit trail and reporting validation
E. All of the Above

A. Account, Contact, and Opportunity creation validation
B. Ownership and role assignment validation
C. Email notification and workflow trigger validation
D. Audit trail and reporting validation
E. All of the Above

Sources

E. All of the Above

Download the Test Strategy Document

Sources

Test Strategy Document – Salesforce Aviation Sales Project

Generated Date: 10-May-2026

1. Project Overview

This document defines the Test Strategy for the Salesforce Aviation Sales Project focused on the Lead Conversion Process. The strategy covers validation of lead qualification, conversion workflows, duplicate handling, integration points, security validations, and post-conversion activities.

2. Scope of Testing

In Scope:
- Lead creation and qualification using BANT criteria
- Lead conversion workflow
- Account, Contact, and Opportunity creation
- Validation rules and duplicate management
- Workflow notifications and audit validation
- Security and role-based access validations

Out of Scope:
- Third-party ERP integrations
- Legacy CRM migration testing
- Performance testing beyond standard Salesforce limits

3. Business Process Covered

Lead Source: None specified by business stakeholders.
Lead Qualification: BANT (Budget, Authority, Need, Timeline).
Conversion Workflow: Lead → Account → Contact → Opportunity.
Duplicate Handling: Manual duplicate verification by sales team.
Post Conversion Validation: Account, Contact, Opportunity, ownership, workflow notifications, reporting, and audit validations.

4. Testing Objectives

3. Business Process Covered

Lead Source: None specified by business stakeholders.
Lead Qualification: BANT (Budget, Authority, Need, Timeline).
Conversion Workflow: Lead → Account → Contact → Opportunity.
Duplicate Handling: Manual duplicate verification by sales team.
Post Conversion Validation: Account, Contact, Opportunity, ownership, workflow notifications, reporting, and audit validations.

4. Testing Objectives

- Validate accurate lead qualification and conversion.
- Ensure successful Account, Contact, and Opportunity creation.
- Verify duplicate prevention and manual review process.
- Validate notifications, approvals, workflows, and reports.
- Ensure data integrity and security compliance.

5. Test Types

- Functional Testing
- Regression Testing
- System Integration Testing
- User Acceptance Testing (UAT)
- Security Testing
- Data Validation Testing

6. Test Environment

Environment: Salesforce Sandbox